

GF4 — Week 2: Pairwise Matching and Epipolar Geometry

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1. Brief Theory

The Epipolar constraint : For corresponding points $\mathbf{x}$, $\mathbf{x}'$ in two images, $\mathbf{x}'^T F \mathbf{x} = 0$, where F is the fundamental matrix. This maps the projection of the ray normal to the plane of the first image to the epipolar line in the second image, on which the corresponding point $\mathbf{x}'$ must lie, reducing the correspondence search from 2-D to 1-D.

The Fundamental matrix : F (3×3 , rank 2, 7 degrees of freedom) encodes the complete projective geometry between two uncalibrated views: it maps any point in one image to its epipolar line in the other, capturing the relative pose and the epipoles through its null spaces.

Using RANSAC for estimating F : Feature matches can often contain outliers (false correspondences). RANSAC iteratively samples minimal 8-point subsets (7-DOF) to fit F and counts inliers; the hypothesis with most inlier support is retained, making the estimate robust to the high outlier fractions typical in raw nearest-neighbour matches.

Why having many matches does not necessarily lead to good reconstruction. Despite a high number of matches, geometrically inconsistent matches or matches that lack sufficient parallax can lead to poor F estimation and inaccurate epipolar geometry. Such cases can include matches that are all clustered in a small region of the image, which may not provide enough parallax for accurate triangulation, or matches that arise from repetitive texture with false correspondences. It is the quality rather than the sheer count of matches that guarantee geometric diversity or epipolar accuracy.

2. Feature and Matching Results



Figure 1: Keypoint visualisations: image 1 (left) and image 2 (right) of the good pair.

We observe that features are highly concentrated on the body of the plush dog due to the high-contrast, repetitive texture, which provides many distinctive keypoints that allows SIFT to reliably detect across scales. In contrast, the white mattress surface is nearly featureless, with its uniform albedo and low-gradient regions yielding almost no detections. The dark background (walls, furniture edges) contributes a small number of large-scale keypoints at structural boundaries.

Table 1: Matching summary for the "good pair".

Metric	Value
Keypoints in image 1	1567
Keypoints in image 2	1277
Raw (1-NN) matches	1567
Filtered matches (Lowe ratio 0.75)	94
RANSAC inliers	53
Inlier ratio	0.564
Median epipolar error, all filtered (px)	0.556
Median epipolar error, RANSAC inliers (px)	0.242



Figure 2: Left: all 1567 raw nearest-neighbour matches. Right: 94 Lowe-filtered matches (ratio = 0.75).

The filtered set is dominated by the plush-body correspondences, which are largely correct. However, a few matches cross structurally similar textured rows that are replicated along the plush dog's body, as these locally similar patch neighbourhoods yield near-equal nearest/second-nearest descriptor distances just below the 0.75 threshold.

3. Fundamental Matrix and Epipolar Geometry

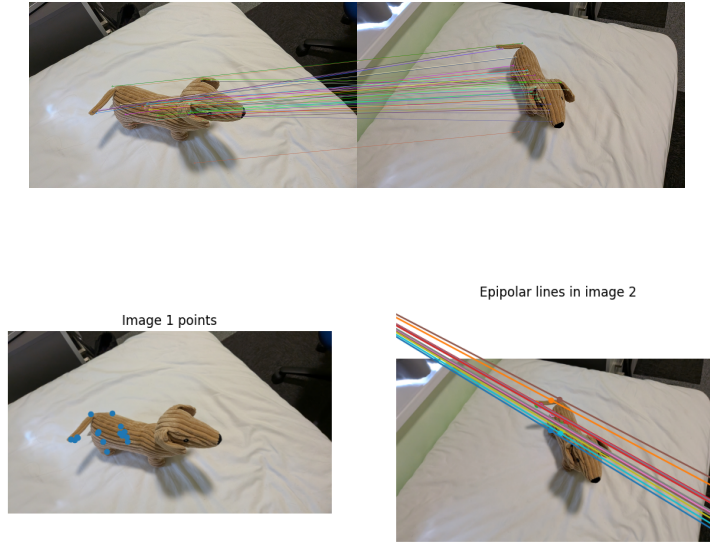


Figure 3: Top: RANSAC inlier matches. Bottom: epipolar-line visualisation (selected inlier points in image 1, corresponding epipolar lines in image 2).

Epipolar lines in image 2 converge toward a single epipole which is consistent with the camera motion, confirming a well-estimated F . RANSAC retained 53 inliers (inlier ratio 0.564) from 94 filtered matches. The median epipolar error for RANSAC inliers is 0.242 px and for all 94 filtered matches 0.556 px which are both sub-pixel, suggesting that the recovered geometry is accurate.

4. Good Pair vs. Bad Pair

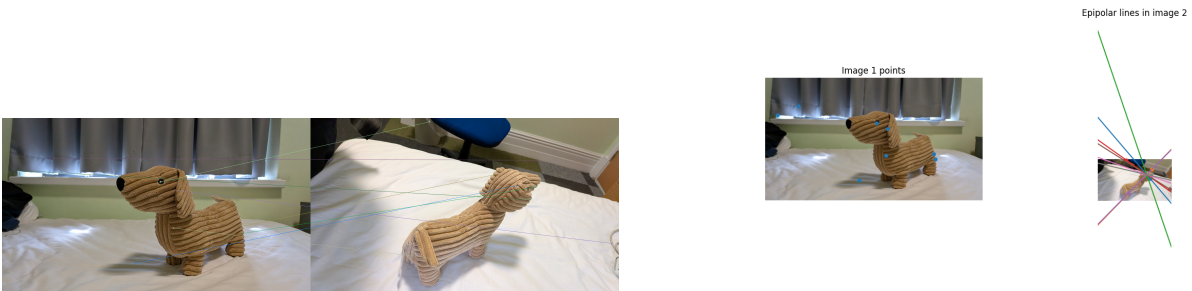


Figure 4: Top: RANSAC inlier matches. Bottom: epipolar-line visualisation (selected inlier points in image 1, corresponding epipolar lines in image 2).

Table 2: Comparison: good pair vs. bad pair

Metric	Good pair	Bad pair
Keypoints in image 1	1567	2535
Keypoints in image 2	1277	2259
Raw matches	1567	2535
Filtered matches (ratio 0.75)	94	17
RANSAC inliers	53	8
Inlier ratio	0.564	0.471
Median epipolar error, all filtered (px)	0.556	20.43
Median epipolar error, RANSAC inliers (px)	0.242	$\approx 0^*$
Main observed failure	None	<9 true inliers; F unreliable

*Near-zero inlier error with 8 points is an artefact: RANSAC overfits to the minimal set (see text).

The bad pair images show the same object in the same position, but from completely different viewpoints, leading to low overlap and a drastically different background and illumination. This causes SIFT to detect many keypoints on the background in image 2 that have no true correspondences in image 1, resulting in a large number of false matches. The nearest and second-nearest descriptor distances are near-equal for these false matches, so 99.3% fail the ratio test (only 17/2535 survive). With merely 8 RANSAC inliers, the estimate can be considered degenerate the near-zero reported inlier error results from RANSAC overfitting rather than correct geometry, as confirmed by the 20.43 px median error over all 17 filtered matches.

5. Targeted Sensitivity Analysis (pair mode only)

Table 3: Sensitivity on the good pair. Study 1: Lowe ratio (RANSAC thr. = 1.0 px fixed). Study 2: RANSAC threshold (ratio = 0.75 fixed).

Study	Setting	Filtered	Inliers	Inlier ratio	Med. epi. (px)
1	ratio = 0.75	94	53	0.564	0.556 (all filt.)
	ratio = 0.90	406	39	0.096	30.16 (all filt.)
2	RANSAC threshold = 1.0 px	94	53	0.564	0.242 (inliers)
	RANSAC threshold = 2.0 px	94	59	0.628	0.280 (inliers)

Study 1: ratio = 0.90 admits $4\times$ more matches but inlier ratio decreases from 56 % to 10 % and median epipolar error rises to 30 px — a flood of false matches RANSAC cannot reject. **Study 2:** threshold = 2.0 px gains 6 inliers at worse accuracy (0.280 vs. 0.242 px). We therefore choose the default settings (ratio = 0.75, threshold = 1.0 px) for a balance of inlier quality and quantity.

6. Pairwise Image-Set Analysis

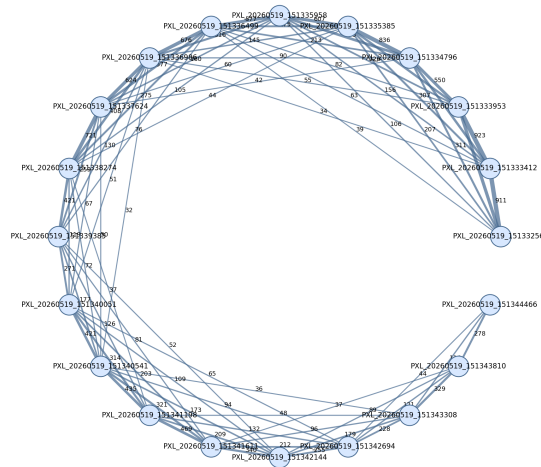


Figure 5: Pairwise match graph for the sub-sampled dog dataset. Node size indicates keypoint count; edge weight shows RANSAC inlier count for pairs which have more than 30 inliers.

The graph forms a near-ring topology, strong edges connect consecutive captures (adjacent nodes), while long-range edges are weak or absent. This is likely due to the camera movement pattern during capture, which involved a

slow, continuous sweep to about the dog, resulting in high overlap between temporally adjacent images and low overlap between non-adjacent ones. Top 3 candidate initial pairs (from `top_initial_pairs.csv`):

Candidate pair	Inliers	Inlier ratio	Med. epi. (px)
...151333412 – ...151333953	923	0.852	0.306
...151332564 – ...151333412	911	0.864	0.208
...151334796 – ...151335385	836	0.868	0.212

All three image pairs are temporally consecutive and exhibit strong geometric consistency, with more than 800 inliers, inlier ratios exceeding 0.85, and sub-pixel median epipolar errors below 0.31 px. These metrics indicate substantial overlap between the images. However, upon closer examination of the image pairs, they seem to be captured from very similar viewpoints, which may limit the parallax and thus the robustness of 3D reconstruction. We therefore choose the subsequent image pairs which also produce high inlier ratios and low epipolar errors, but with slightly more viewpoint variation to provide better geometric constraints for reconstruction:

Candidate pair	Inliers	Inlier ratio	Med. epi. (px)
...151333953 – ...151335385	370	0.749	0.432
...151342144 – ...151343308	179	0.696	0.395
...151340541 – ...151341611	321	0.821	0.407

Preferred initial pair: 151340541–151341611 (highest inlier ratio, lowest median epipolar error of 0.407px), giving the most reliable seed for incremental reconstruction.

7. Group Dynamics

- **Jerry (yhl63):** Setup shared workspace, implemented phases A and B functions such as SIFT keypoint detection, knn matching, Lowe ratio filtering, and RANSAC-based F estimation, as well as the dataset-mode processing.
- **Jashwanth (jp2057):** Implemented phases C and D, including epipolar line visualisation, image-pair analysis and conducted the targeted sensitivity analysis between good and bad pairs.
- **Roksy (ri302):** Implemented phase E of refactoring feature-pair analysis and performing dataset-wide pairwise analysis.

8. Experiment LogW

Implementation steps and group contributions

Shared repository and working environment setup, as well as calling opencv functions for SIFT and RANSAC, were completed by Jerry, followed by implementations of the epipolar line visualisation and image-pair analysis pipeline by Jashwanth. Image datasets were the same as week1, where the dataset with the most succesful reconstruction were used as image pairs for week2. The dataset-wide pairwise analysis was conducted by Roksy, who also generated the pairwise metrics and match graph for the sub-sampled dog dataset.

Debugging record

No major bugs were encountered, but some unexpected results prompted deeper investigation:

Ratio = 0.90 gave fewer RANSAC inliers than ratio = 0.75. Expected: more filtered matches $\Rightarrow$ more inliers. Actual: inlier ratio fell from 56 % to 10 %. Diagnosis: `matches_filtered.png` showed geometrically inconsistent lines; `median_epipolar_error_all` rose to 30 px, confirming a flood of ambiguous false matches that overwhelmed RANSAC. Fix: retain ratio = 0.75.

Bad pair: near-zero RANSAC inlier error despite broken geometry. RANSAC reported almost 0px inlier error, yet `median_epipolar_error_all`=20.43 px. Diagnosis: 8 inliers fitting a 7-DOF model $\Rightarrow$ RANSAC overfits a degenerate subset. The all-matches median is the true diagnostic.

Experiment choices

Good pair: same toy, moderate baseline; chosen because epipolar lines converge and inlier error is sub-pixel.

Bad pair: same toy photographed from widely different viewpoints, chosen to show the ratio-test failure when backgrounds differ.